



BAULKHAM HILLS HIGH SCHOOL

JUNE 2012
YEAR 11 TASK 2

Mathematics

General Instructions

- Reading time – 5 minutes
- Working time – 60 minutes
- Write using black or blue pen
- Board-approved calculators may be used
- Show all necessary working in questions 6 to 14
- Marks may be deducted for careless or badly arranged work

Total marks – 39

Exam consists of 4 pages.

This paper consists of TWO sections.

Section 1 – Page 2

Questions 1-5 (5 marks)

- Attempt Question 1-5

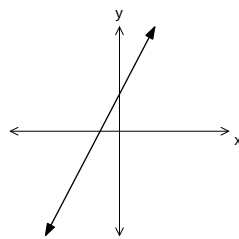
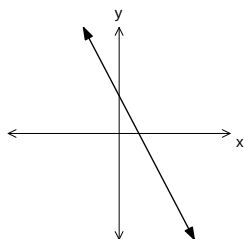
Section II – Pages 3-4 (34 marks)

- Attempt questions 6 to 14

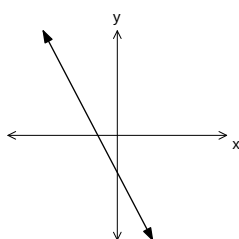
Section I - 5 marks
Attempt questions 1-5

Use the multiple choice answer sheet for question 1-5

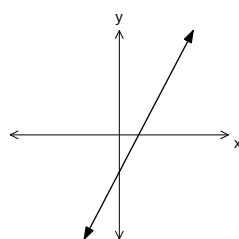
1. Which line could have the equation $y = -2x + 4$?
 (A) (B)



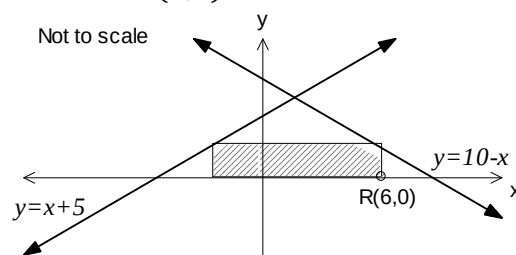
(C)



(D)



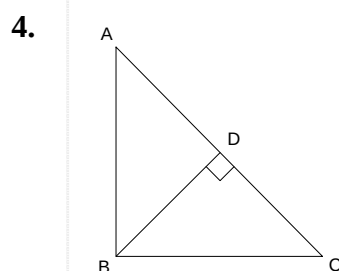
2. If R is the point with co-ordinates $(6,0)$



The area of the shaded rectangle is

- (A) 20 (B) 24 (C) 28 (D) 60

3. If $(2, k)$ lies on the line $x + 2y = 8$ then the value of k is
 (A) 2 (B) 3 (C) 4 (D) 5

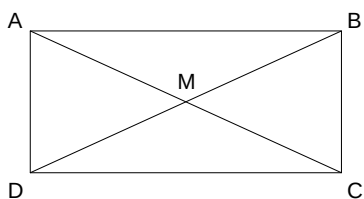


In the diagram, $\triangle ABD$ is similar to $\triangle CBD$

Which of the following is not necessarily true?

- (A) $AB = BC$ (B) $AD = DC$
 (C) $AB \perp BC$ (D) $BD \perp AC$

5. In the diagram, $\triangle AMB \equiv \triangle BMC \equiv \triangle CMD \equiv \triangle DMA$.
 $ABCD$
 (A) is a parallelogram (B) is a rectangle
 (C) is a rhombus (D) cannot be determined



End of Section 1

Section II – Extended Response

Attempt questions 6-14.

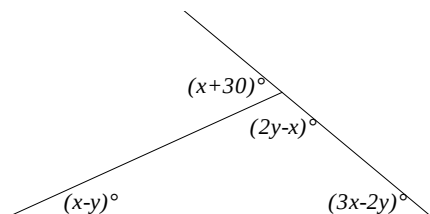
Answer each question on a SEPARATE PAGE. Clearly indicate question number.

Each piece of paper must show your name.

All necessary working should be shown in every question.

6. Find the values of x and y in the figure below. (no reasons required)

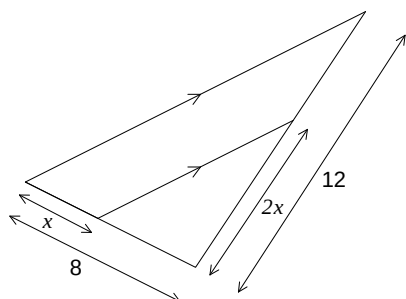
2



- 7.

In the diagram below, find the value of x .
(no reasons required)

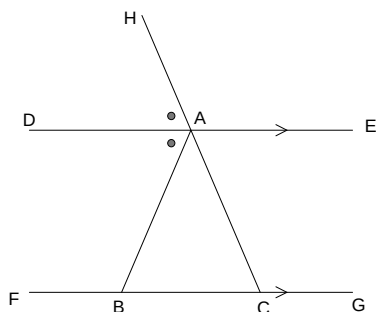
2



- 8.

Prove that $\triangle ABC$ is an isosceles triangle.

3



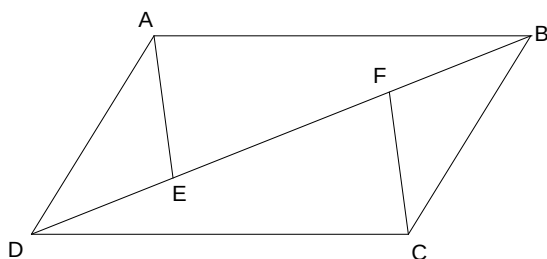
- 9.

$ABCD$ is a parallelogram with $DE = BF$

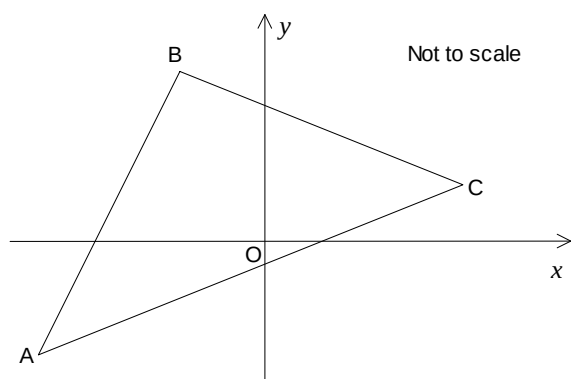
- a) Show that $\triangle AED \equiv \triangle BCF$
b) Prove that AE is parallel to CF

2

2



10.



The diagram shows the points $A(-3, -2)$, $B(-1, 4)$ and $C(5, 2)$

Copy this diagram into your answer booklet.

- Find the gradient of AC
- Point P is the midpoint of AC . Show that the coordinates of P are $(1, 0)$.
Mark P on your diagram.
- Show that the equation of the line perpendicular to AC and passing through the point P is $2x + y - 2 = 0$
- Show that B lies on the line $2x + y - 2 = 0$
- Show that the length of BP is $2\sqrt{5}$ units
- Point P is the midpoint of BD . Find the coordinates of D
- Explain why $ABCD$ is a rhombus
- Find the area of the rhombus $ABCD$

1
1
2
1
1
1
1
2

11. Show that the line $7x - 4y + 5 = 0$ passes through the point of intersection of the lines $3x - 2y + 3 = 0$ and $2x - y + 1 = 0$ without finding the point of intersection of these lines.

3

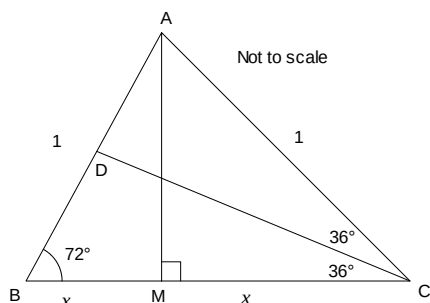
12. Show that the perpendicular distance of the point $(2p, -p)$ from the line $4px + 3py + 5 = 0$ is $\frac{p^2+1}{|p|}$

2

13. The sides of a triangle are $(2x - 1)$ cm, $(x + 1)$ cm and $(11 - x)$ cm.
Show that $2\frac{3}{4} < x < 6\frac{1}{2}$

2

14.



In the diagram, ABC is an isosceles triangle where $\angle ABC = \angle BCA = 72^\circ$,
 $AB = AC = 1$ and $BC = 2x$.

Angle BCA is bisected by CD and angle BAC is bisected by AM which is also the perpendicular bisector of BC .
Copy the diagram in your answer booklet.

- Show that $AD = 2x$
- Show that triangles ABC and CBD are similar.
- Using (b), find the exact value of x
- Hence find the exact value of $\sin 18^\circ$

2
1
2
1

~ End of Exam ~

M-C	1	A
	2	C
	3	D
	4	C
	5	C

$$\begin{aligned}
 6. \quad x + 30 + 2y - 2 &= 180 \\
 2y &= 150 \\
 y &= 75 \quad \checkmark \\
 x + 30 &= x - y + 3x - 2y \\
 30 + 3y &= 3x \\
 3x &= 255 \\
 x &= 85 \quad \checkmark \\
 \therefore x &= 85, y = 75
 \end{aligned}$$

$$\begin{aligned}
 7. \quad \frac{2x}{12} &= \frac{8-x}{8} \quad \checkmark \\
 16x &= 12(8-x) \\
 4x &= 3(8-x) \\
 4x &= 24 - 3x \\
 7x &= 24 \\
 x &= 3\frac{3}{7} \quad \checkmark
 \end{aligned}$$

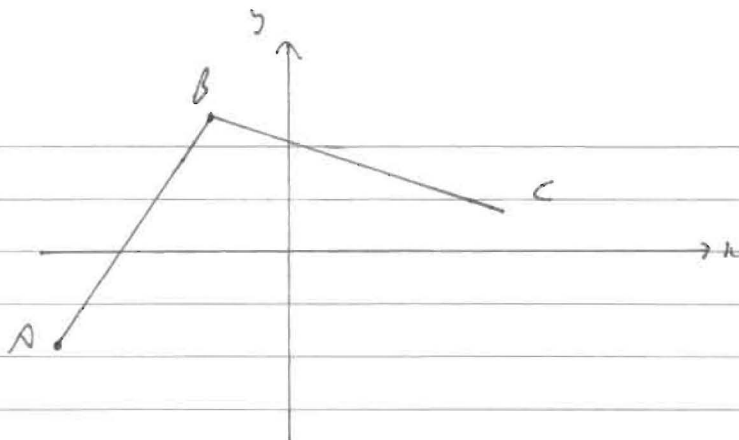
$$\begin{aligned}
 8. \quad \angle ACB &= \angle HAD \quad (\text{corresponding angles, } DE \parallel FG) \quad \checkmark \\
 \angle ABC &= \angle DAB \quad (\text{alternate angles, } DE \parallel FG) \quad \checkmark \\
 \therefore \angle ABC &= \angle ACB \quad (\angle HAD = \angle DAB) \quad \checkmark
 \end{aligned}$$

9.a) In Δ 's AED and CFB

$$\begin{aligned}
 AD &= BC \quad (\text{opposite sides of a parallelogram}) \\
 DE &= BF \quad (\text{given}) \\
 \angle ADE &= \angle FBC \quad (\text{alternate angles, } AD \parallel BC) \\
 \therefore \Delta AED &\equiv \Delta BCF \quad (\text{SAS}) \quad \checkmark
 \end{aligned}$$

$$\begin{aligned}
 b) \quad \angle BFC &= \angle AED \quad (\text{matching angles in congruent } \Delta\text{'s AED, CFB}) \\
 \angle CFE &= 180^\circ - \angle BFC \quad (\text{angle sum on straight line EB}) \\
 \angle AEF &= 180^\circ - \angle AED \quad (\text{angle sum on straight line EB}) \quad \checkmark \\
 \therefore \angle AEF &= \angle EFC \\
 \therefore AE &\parallel FC \quad (\text{alternate angles are equal}) \quad \checkmark
 \end{aligned}$$

10.



$$\begin{aligned} \text{a) } m_{AC} &= \frac{2 - -2}{5 - -3} \\ &= \frac{4}{8} \\ &= \frac{1}{2} \quad \checkmark \end{aligned}$$

$$\begin{aligned} \text{b) } \text{Midpoint} &= \left(\frac{-3+5}{2}, \frac{-2+2}{2} \right) \\ &= (1, 0) \quad \checkmark \end{aligned}$$

$$\begin{aligned} \text{c) } m_{\perp AC} &= -2 \quad \checkmark \\ y - 0 &= -2(x - 1) \\ y &= -2x + 2 \\ \therefore 2x + y - 2 &= 0 \quad \text{as required} \quad \checkmark \end{aligned}$$

$$\begin{aligned} \text{d) } LHS &= 2x + y - 2 \\ &= 2(-1) + 4 - 2 \\ &= 0 \\ &= RHS \quad \checkmark \\ \therefore B &\text{ lies on } 2x + y - 2 = 0 \end{aligned}$$

$$\begin{aligned} \text{e) } BP &= \sqrt{(-1-1)^2 + (4-0)^2} \\ &= \sqrt{(-2)^2 + 4^2} \\ &= \sqrt{20} \\ &= 2\sqrt{5} \quad \checkmark \end{aligned}$$

$$\begin{aligned} \text{f) } 1 &= \frac{-1+x}{2} & 0 &= \frac{4+y}{2} \\ 2 &= -1+x & 0 &= 4+y \\ x &= 3 & y &= -4 \\ \therefore D &\text{ is } (3, -4) \quad \checkmark \end{aligned}$$

g) Diagonals bisect each other at 90° ✓
(ie P is midpoint of diagonals BD and AC).

$$10. h) CP = \sqrt{(5-1)^2 + (2-0)^2}$$

$$= \sqrt{4^2 + 2^2}$$

$$= 2\sqrt{5}$$

$$\text{Area} = \left(\frac{2\sqrt{5} \times 2\sqrt{5}}{2} \right) \times 4$$

$$\therefore \text{Area} = 40 \text{ units}^2$$

$$11. \text{ let } 3x - 2y + 3 + k(2x - y + 1) = 0 \text{ where } k \text{ is a real no.}$$

$$(3+2k)x - (2+k)y + 3+k = 0$$

$$\text{gradient} = \frac{3+2k}{2+k}$$

$$\text{Now gradient of } 7x - 4y + 5 = 0 \text{ is } m = \frac{7}{4}$$

$$\therefore \frac{3+2k}{2+k} = \frac{7}{4}$$

$$12 + 8k = 14 + 7k$$

$$k = 2$$

$\therefore$ Equation is

$$(3+4)x - (2+2)y + 3+2 = 0$$

$$7x - 4y + 5 = 0 \text{ as req'd}$$

$$12. \text{ Perp. dist} = \left| \frac{4p \times 2p + 3p \times -p + 5}{\sqrt{(4p)^2 + (3p)^2}} \right|$$

$$= \left| \frac{8p^2 - 3p^2 + 5}{\sqrt{16p^2 + 9p^2}} \right|$$

$$= \frac{|5p^2 + 5|}{\sqrt{25p^2}}$$

$$= \frac{5|p^2 + 1|}{5|p|}$$

$$\therefore \text{Perp distance} = \frac{p^2 + 1}{|p|}$$

13. Since the sum of the two shorter sides of a triangle are more than the longest side, then

$$2x-1 + x+1 > 11-x \text{ and } 11-x + x+1 > 2x-1$$

$$3x > 11-x$$

$$12 > 2x-1$$

$$4x > 11$$

$$2x < 13$$

$$x > \frac{11}{4}$$

$$x < 6\frac{1}{2}$$

$$x > 2\frac{3}{4}$$

Note $11-x+2x-1 > x+1$ & This is since if $x > 2\frac{3}{4}$
 $x+10 > x+1$ then $x+1$ is not the longest
 $10 > 1$ side
 $\therefore 2\frac{3}{4} < x < 6\frac{1}{2}$ ✓

14. a) $\angle COB = 72^\circ$ (angle sum of $\triangle BCO$)

$CO = BC$ (equal sides opposite equal angles in $\triangle BCO$) ✓

$$\therefore CO = 2x \text{ (BC = 2x)}$$

$\angle OAC = 36^\circ$ (angle sum of $\triangle BOC$)

$\therefore CO = AO$ (equal sides opposite equal angles in $\triangle ACO$) ✓

$$\therefore AO = 2x \text{ (CO = 2x)}$$

b) In $\triangle ABC$ and $\triangle CBD$

$\angle ABC$ is common

$$\angle BOC = \angle ACB = 72^\circ \text{ (from (a))}$$

$\therefore \triangle ABC \parallel \triangle CBD$ (two pairs of angles are equal) ✓

c) $\frac{CB}{AC} = \frac{BO}{BC}$ (matching sides in similar $\triangle ABC$ & $\triangle CBO$)

$$\frac{2x}{1} = \frac{1-2x}{2x}$$

$$4x^2 = 1-2x$$

$$4x^2 + 2x - 1 = 0$$

$$x = \frac{-2 \pm \sqrt{4 - 4 \times 4 \times -1}}{2 \times 4}$$

$$x = \frac{-2 \pm \sqrt{20}}{8}$$

$$x = \frac{-2 \pm 2\sqrt{5}}{8}$$

$$x = \frac{-1 \pm \sqrt{5}}{4}$$

But $x > 0$, $\therefore x = \frac{\sqrt{5}-1}{4}$ ✓

$\angle MAB = 18^\circ$ (subtracts $\angle BAC$)

$\sin 18^\circ = \frac{x}{1}$ ✓

$\sin 18^\circ = \frac{\sqrt{5}-1}{4}$ ✓